

Abstract

We present the Holographic Theory of State Persistence (THPE), a phenomenological extension of holographic dark energy in which the dark energy density is promoted to a field $\Phi(z) = \Phi_0[1 + \alpha f_{\text{SFR}}(z) + \beta g_{\text{struct}}(z) + \gamma h_{\text{ent}}(z)]$ tied to three tracers of informational complexity: star formation, large-scale structure and horizon entropy. The model recovers Λ CDM exactly for $\alpha = \beta = \gamma = 0$ and predicts a crossing of $w(z) = -1$ synchronised with the star formation history. We report two complementary results. (i) Confronting the model with DESI DR2 baryon acoustic oscillations (13 measurements with per-tracer correlations), the Pantheon+ Type Ia supernova compilation (1588 objects with the full statistical-plus-systematic covariance and the absolute calibration analytically marginalised) and the compressed CMB distance priors from Planck 2018, nested sampling yields $\ln B = -7.4 \pm 0.2$ in favour of Λ CDM against the minimal single-tracer variant and $\ln B = -14.8 \pm 0.2$ against the three-parameter variant: strong evidence on the Jeffreys scale in all cases. The horizon-entropy term is excluded by physicality ($|\gamma| \lesssim 10^{-3}$) and the final bounds are $\alpha < 0.008$ and $\beta < 0.07$ (68% CL), so that any informational contribution to the holographic pressure lies below $\sim 1\%$ of the dark energy density. (ii) Investigating whether $\Phi(z)$ can be derived rather than postulated, we find that a covariant action with non-minimal coupling does generate $\square\Pi = I$ with the source fixed by the coupling, and that the construction passes ghost, degrees-of-freedom and conservation tests; however, the Cassini bound on the scalar sector is ~ 400 times more restrictive than cosmology, and the known screening mechanisms lead to theories already present in the literature. We conclude that the local scalar–tensor formulation of the THPE provides no independent phenomenological content relative to the Horndeski theories surviving GW170817. We additionally show that identifying the Compton time $\tau_c = \hbar/Mc^2$ as a universal state-compression rate is refuted by demonstrated quantum coherence by 14 to 51 orders of magnitude. The full pipeline, its coherence controls and the complete correction history are made public. The prediction of a $w = -1$ crossing tied to star formation remains registered as a falsifiable test for the coming decade.

The Holographic Theory of State Persistence confronted with DESI DR2, Pantheon+ and the CMB: observational bounds and theoretical localisation of an informational extension of holographic dark energy

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1 Introduction

The nature of dark energy remains unexplained. Recent DESI results have reopened the possibility that its equation of state evolves with time [1, 2], motivating the study of dynamical extensions of Λ CDM.

The THPE proposes one such extension: that the dark energy density be tied to the informational content of the universe — the complexity generated by star formation, structure growth and horizon entropy. The underlying motivation is that a universe which produces structure might leave a measurable imprint on its own expansion.

This paper does not defend that hypothesis: it tests it. An earlier version of this work presented the model as a proposal with qualitative compatibility. The present version reports the complete outcome of two consecutive research programmes — one observational and one mathematical — both with acceptance criteria fixed in advance, and both yielding negative results which we publish in full. We regard the delimitation of the viable parameter space of a hypothesis as a legitimate scientific product.

The paper is organised as follows. Section 2 presents the model. Section 3 describes the data and the inference methodology, including the coherence controls. Section 4 reports the observational results. Section 5 refutes the local Compton scale. Section 6 reports the attempt to derive $\Phi(z)$ from an action principle and localises the resulting theory. Sections 7–9 discuss the limitations and the future testability of the prediction.

2 Theoretical framework

2.1 Formulation

The THPE promotes the dark energy density to

$$\Phi(z) = \Phi_0 [1 + \alpha f_{\text{SFR}}(z) + \beta g_{\text{struct}}(z) + \gamma h_{\text{ent}}(z)], \quad (1)$$

where f_{SFR} is the normalised cosmic star formation rate [3], g_{struct} a normalised large-scale structure density, and h_{ent} the horizon entropy per comoving volume. The Friedmann equation reads

$$H^2(z) = H_0^2 [\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Phi(z)], \quad (2)$$

and the effective equation of state is

$$w(z) = -1 + \frac{1+z}{3\Phi} \frac{d\Phi}{dz}. \quad (3)$$

For $\alpha = \beta = \gamma = 0$, Eq. (1) gives $\Phi = \Phi_0$ and Λ CDM is recovered exactly, with $\Phi_0 = \Omega_\Lambda$. The characteristic prediction is a crossing of $w(z) = -1$ whose chronology follows the peak of star formation at $z \simeq 1.9$.

2.2 Relation to previous work

The Hilbert-space layer structure separated by the Planck time coincides with the Holographic Space-Time framework of Banks and Fischler [4]; we state this without claiming originality on that point. The proposed contribution was the explicit link between the holographic pressure and observable tracers of complexity. Section 6.7 documents a direct precedent for the underlying intuition in the causal set literature.

2.3 Status after this work

Section 6 shows that the local scalar–tensor version of Eq. (1) is localised within an already studied space of theories. Accordingly, Eq. (1) is to be understood in this paper as an observationally bounded *phenomenological parametrisation*, not as a proposal of new fundamental physics.

3 Data and methodology

3.1 Data

We use three datasets.

(i) *DESI DR2 BAO* [2]: 13 measurements — D_V/r_d for the BGS tracer, and the pairs $(D_M/r_d, D_H/r_d)$ with their correlation coefficients for LRG1, LRG2, LRG3+ELG1, ELG2, QSO and Ly α , spanning $0.295 \leq z_{\text{eff}} \leq 2.330$. The transcription of the data vector was verified against two independent sources.

(ii) *Pantheon+* [5]: 1588 Type Ia supernovae with $z > 0.01$ and the full statistical-plus-systematic covariance matrix.

(iii) *Compressed CMB distance priors* from Planck 2018 [6, 7]: the shift parameter R and the acoustic scale ℓ_A with their correlation, self-calibrated to the fiducial background of the pipeline. The common modelling offset ($\sim 0.1\%$, arising from the treatment of neutrinos and from the numerical evaluation of r_s) is identical for all models and cancels by construction; this choice is stated explicitly.

3.2 Likelihoods

For BAO we adopt a Gaussian likelihood with a block-diagonal covariance per tracer. For supernovae we marginalise analytically over a constant magnitude offset,

$$\chi_{\text{marg}}^2 = \Delta^T C^{-1} \Delta - \frac{(\Delta^T C^{-1} \mathbf{1})^2}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad (4)$$

where $\Delta = \mu_{\text{obs}} - \mu_{\text{th}}$. This renders the result independent of the SH0ES versus Planck absolute calibration. For the CMB we use a two-dimensional Gaussian in (R, ℓ_A) . The background is fixed to Planck 2018 values ($H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$); this limitation is discussed in Sec. 7.

3.3 Inference

Priors: Φ_0 log-uniform on $[10^{-3}, 2]$; α and β uniform on $[0, 10]$; γ uniform on $[0, 5]$ (negative values imply $H^2 < 0$ at $z \lesssim 0.01$, where supernovae exist, and are excluded from the prior); $\Phi(z) > 0$ is enforced on $0.011 \leq z \leq 4.5$. Evidences are computed with nested sampling (DYNESTY [8], $n_{\text{live}} = 800$, slice sampling).

We report a methodological warning of general applicability: ensemble samplers of the Goodman–Weare type [9] do *not* converge on this posterior. The extended Φ_0 – α – β degeneracy produces Gelman–Rubin statistics between 3 and 8 even with 64 walkers and 20000 steps. Nested sampling converges and additionally provides the evidence.

3.4 Coherence controls

Two automatic controls were built into the pipeline. (i) *Reference-model coherence*: χ^2 of Λ CDM against each data vector is computed at start-up, and execution aborts if it is anomalous. (ii) *Nesting condition*: $\ln \mathcal{L}_{\text{max}}(\text{THPE}) \geq \ln \mathcal{L}_{\text{max}}(\Lambda\text{CDM})$ must hold, since the models are nested.

These controls detected and discarded three invalid results during the campaign. The most instructive was a spurious $\ln B = +290$ *in favour* of the THPE, caused by a truncation in the numerical evaluation of the sound horizon ($r_s = 139.0$ instead of 144.4 Mpc). We report this explicitly because it illustrates that, in Bayesian model comparison, a coherence control on the *reference* model is indispensable: an anomalous χ^2 for the model known to work signals a bookkeeping error, not a discovery.

4 Observational results

4.1 Evidences

With the full geometric dataset (DESI DR2 BAO + Pantheon+ + CMB distance priors) we obtain

$$\begin{aligned} \Lambda\text{CDM (1 par.)} : \quad \ln Z &= -740.13 \pm 0.11, \\ \text{THPE } \alpha\text{-only (2 par.)} : \quad \ln Z &= -747.56 \pm 0.16, \\ \text{THPE } (\gamma = 0, 3 \text{ par.}) : \quad \ln Z &= -754.92 \pm 0.19, \end{aligned} \tag{5}$$

so that

$$\ln B(\alpha\text{-only}/\Lambda\text{CDM}) = -7.43 \pm 0.19, \tag{6}$$

$$\ln B(3\text{-par.}/\Lambda\text{CDM}) = -14.79 \pm 0.22. \tag{7}$$

The nesting control is satisfied: the three models share $\ln \mathcal{L}_{\text{max}} = -733.50$. The best-fitting THPE configuration coincides exactly with Λ CDM ($\Delta\chi^2 = 0.0$). For Λ CDM the posterior gives $\Phi_0 = 0.6883$, consistent with the Planck value.

4.2 Robustness chain

The result is stable across dataset combinations:

Dataset	Statistic	Value
DR1 BAO+SNe (MCMC)	ΔAIC	+10.5
DR2 BAO+SNe	$\ln B$ (3-par.)	−11.1
DR2 BAO+SNe	$\ln B$ (α -only)	−4.9
DR2+CMB	$\ln B$ (3-par.)	−14.8
DR2+CMB	$\ln B$ (α -only)	−7.4

Adding the CMB anchor *worsens* the case for the THPE. Without the anchor, Λ CDM prefers $\Phi_0 = 0.724 > 0.685$, breaking flatness: the mild ($\sim 2.3\sigma$) DR2–Planck tension projects onto the *amplitude* of dark energy, not onto its functional form, and neither f_{SFR} nor g_{struct} has the required shape.

4.3 Parameter bounds

At 68% credibility,

$$\alpha < 0.008, \quad \beta < 0.07, \quad |\gamma| \lesssim 10^{-3}. \quad (8)$$

The horizon-entropy term is excluded as formulated: with the adopted normalisation h_{ent} diverges as z^{-3} for $z \rightarrow 0$, so positivity of H^2 where supernovae exist, together with the nearby supernova distances, confines γ to zero. Any informational contribution to the holographic pressure therefore lies below $\sim 1\%$ of the dark energy density.

The degeneracy direction is well determined, $\Phi_0(1 + \alpha + \beta) \simeq \Omega_\Lambda$: the fit distributes the same dark energy amplitude among three parameters.

5 Refutation of the local Compton scale

An earlier version of the THPE identified the Compton time $\tau_c = \hbar/Mc^2$ as a universal rate of state compression. Comparing that scale with experimentally demonstrated quantum coherence times — from electron interferometry through molecular interferometry [10] to the 40 kg mirrors of LIGO — the observed coherence exceeds the prediction by 14 to 51 orders of magnitude across every mass range ever probed. The identification is therefore refuted.

The only surviving reading of local persistence is strictly relational (the state $|\psi\rangle$ persists, not a classical trajectory; compression occurs only through environmental interaction), which is consistent but operationally equivalent to standard decoherence and thus carries no additional empirical content. Diósi–Penrose-type scales [11, 12], $\tau \sim \hbar R/Gm^2$, remain compatible with all observations and testable with current optomechanics [13]; their exploration lies outside the scope of this work.

6 Can $\Phi(z)$ be derived?

6.1 Set-up

We investigated whether Eq. (1) can be obtained from a variational principle rather than postulated. The success criterion, fixed in advance, was to obtain $\square\Pi = I$ as a *consequence*, with the source I deduced rather than chosen, while passing ghost, degrees-of-freedom and conservation tests.

6.2 Minimal case

Consider the simplest possible non-minimal coupling,

$$S = \int d^4x \sqrt{-g} \left[\frac{F(\Pi)R}{2\kappa} - \frac{\omega}{2}(\partial\Pi)^2 - V(\Pi) \right] + S_m, \quad (9)$$

with $F(\Pi) = 1 + 2\kappa\xi\Pi$. Variation with respect to Π gives

$$\omega \square\Pi + \xi R - V'(\Pi) = 0 \quad \implies \quad \square\Pi = \frac{V'(\Pi) - \xi R}{\omega}. \quad (10)$$

The target equation therefore emerges, and the source is *not* free: $I \propto R$ in vacuum and $I \propto T$ (the trace of the energy-momentum tensor) in the presence of matter, via the trace of the modified Einstein equations, yielding the canonical Brans–Dicke form $(2\omega_{\text{BD}} + 3)\square\Pi = \kappa T$.

All three consistency tests are passed. In the Einstein frame the kinetic coefficient is

$$K_E = \frac{\omega}{F} + \frac{3}{2\kappa} \left(\frac{F'}{F} \right)^2 = \frac{\omega F + 6\kappa \xi^2}{F^2} > 0 \quad \text{for } \omega > 0, F > 0, \quad (11)$$

so there is no ghost provided $F > 0$ (the region $F < 0$ must be excluded by a physicality prior, in exact analogy with the prior used in the cosmological analysis). The theory propagates three degrees of freedom (two tensor, one scalar) with $c_{\text{gw}} = c$ exactly, compatible with GW170817 [14]. Energy–momentum conservation holds identically because matter couples minimally to $g_{\mu\nu}$ in the Jordan frame.

6.3 The bottleneck is local, not cosmological

The extra scalar mode is precisely that of Brans–Dicke [15], with $\omega_{\text{BD}} = \omega F / (F')^2$. The Cassini bound $\omega_{\text{BD}} > 4 \times 10^4$ [16] implies

$$|\xi| < 2.5 \times 10^{-3} \kappa^{-1} \implies \left| \frac{\delta H}{H} \right| < 2.5 \times 10^{-5}. \quad (12)$$

The Solar System limit is thus roughly *400 times more restrictive* than the cosmological bounds of Eq. (8). The minimal formulation is consistent but indistinguishable from Λ CDM by construction — and would remain so for any future cosmological dataset.

6.4 Screening mechanisms

A field-dependent kinetic term $Z(\Pi)$ cannot produce screening. The redefinition

$$\frac{d\phi}{d\Pi} = \sqrt{Z(\Pi)} \quad (13)$$

removes any $Z(\Pi) > 0$, returning the theory of Eq. (9) written in different field-space coordinates; the effective coupling is $F'/\sqrt{Z}$, a fixed number rather than a function of the environment. Genuine screening requires dependence on the local density (chameleon [17], symmetron) or on the derivatives $X = -(\partial\Pi)^2/2$ (K-mouflage, Vainshtein). The former are subject to the no-go theorem of Wang, Hui and Khoury [18], which forbids self-acceleration.

6.5 Vainshtein and Horndeski

After GW170817, the viable Horndeski Lagrangian reduces to

$$\mathcal{L} = G_2(\Pi, X) + G_3(\Pi, X) \square\Pi + G_4(\Pi)R, \quad G_5 = 0, \quad (14)$$

and Vainshtein screening [19] from the G_3 term does resolve Eq. (12): the solar Vainshtein radius is $r_V = (r_s L_c^2)^{1/3} \simeq 3.8 \times 10^{18} \text{ m} \simeq 124 \text{ pc}$, suppressing the fifth force by $\sim 10^{-9}$ at Neptune’s orbit. A cosmological window between 10^{-3} and 10^{-2} remains, bounded above by Eq. (8) and further restricted by graviton decay into dark energy [21] and by gradient-stability requirements [22].

However, the local formulation analysed for Π is mathematically equivalent to a subclass of Galileon/Horndeski theories [20] within the adopted hypotheses. The retarded solution of Eq. (10),

$$\Pi(x) = \int G_{\text{ret}}(x, y) I(y) \sqrt{-g} d^4y, \quad (15)$$

is a mathematical property of *any* sourced field — the retarded Liénard–Wiechert potentials have possessed such “causal memory” since 1898 — and adds neither degrees of freedom nor distinctive predictions.

6.6 The Planck scale: the real limit of the programme

The original intuition placed state persistence at the Planck scale, $t_P = \sqrt{\hbar G/c^5} \simeq 5.4 \times 10^{-44}$ s, with $\ell_P = ct_P$. Yet the Planck time appears nowhere in the formulation analysed above: Eq. (10) is a continuous wave equation and Eq. (15) integrates over a continuous past light cone. Formalising Π as a classical field with a covariant action removes the discreteness by construction.

In this formulation gravity influences Π in two ways, neither involving the Planck scale: as a *source*, through R (and through T in the presence of matter) in Eq. (10); and as a *response*, through the $F(\Pi)R$ term, which makes the effective gravitational constant depend on Π and is precisely what Eq. (12) constrains. Both effects are macroscopic and classical.

This yields a sharper reading of our result: the local formulation does not fail because of the Cassini bound — Vainshtein screening resolves that — but because, on being made rigorous, it *loses the ingredient that made it distinctive*. A classical field theory cannot represent Planck-scale granularity, and without it Π is one more scalar field. If state persistence is genuinely a Planck-scale phenomenon, its description requires a quantum gravity framework rather than an effective field theory. We regard this — rather than the observational constraints — as the real limit of the theoretical programme presented here.

6.7 Precedent in the literature: Everpresent Lambda

It is natural to ask whether that route has already been taken. It has. In causal set theory, Lorentz covariance imposes a Poisson relation between the spacetime volume of a region and the number N of discrete elements it contains. From this Sorkin [24, 25] deduced a fluctuating cosmological constant with $\delta\Lambda \sim 1/\sqrt{V} \sim 1/\sqrt{N}$: the accumulation of discrete Planck-scale elements produces, in the macroscopic limit, dynamical dark energy. The effect is explicitly described as a residual, non-local quantum consequence of the underlying discreteness [26, 27]. The prediction predates the 1998 discovery of cosmic acceleration and its order of magnitude agrees with the observed value without fine-tuning.

The correspondence with the motivation of the present work is close and we state it explicitly: both start from Planck-scale discrete structure, both posit that its accumulation affects the expansion, both produce dynamical dark energy, and both are non-local by construction. The difference is formal status: Everpresent Lambda *derives* the fluctuation from a quantum gravity theory, whereas the THPE *postulated* the functional form of Eq. (1) from phenomenological tracers.

That programme is likewise constrained, and by the same observable: since the everpresent Λ originates in past light cones, regions of the last-scattering surface separated by more than a causal horizon would carry uncorrelated values, producing CMB anisotropies; the resulting bound is too small to account for late-time acceleration [28]. The conclusion of Sec. 6.5 is therefore reinforced rather than weakened: the Planck route is not free territory left unexplored, but ground already covered, formalised and observationally restricted.

7 Discussion and limitations

Three lessons follow. First, the functional form of $\Phi(z)$ is not what the data require: the DR2–Planck tension projects onto the amplitude of dark energy rather than its shape. Second, the Φ_0 – α – β degeneracy is structural; separating the contributions would require growth observables such as $f\sigma_8$ rather than purely geometric ones. Third, h_{ent} must either be reformulated from theory, regularising its $z \rightarrow 0$ divergence, or dropped.

Limitations are stated explicitly. H_0 and Ω_m are fixed to Planck values; marginalising over them would broaden the bounds of Eq. (8) without changing the sign of Eqs. (6)–(7), which is dominated by the Occam penalty. We do not use the full CMB likelihood, nor structure

growth. The CMB priors are self-calibrated as described in Sec. 3. The theoretical analysis is restricted to local, single-field, second-order actions within the space of Eq. (14); genuinely non-local formulations [23], history functionals, and multi-field sectors are *not* refuted here — they were not examined.

8 Conclusions

We have promoted the THPE from proposal to tested hypothesis on two independent fronts.

Observationally, with the best available public geometric datasets and explicit validity controls, the evidence strongly favours Λ CDM ($\ln B$ from -7.4 to -14.8), excludes the horizon-entropy term, and bounds informational contributions below $\sim 1\%$ of the dark energy density.

Theoretically, the target equation can be derived from an action principle, but the resulting local formulation is localised within an already studied space of theories and provides no independent phenomenology. The Compton scale is refuted as a physical compression rate, and the Planck-scale route proves already covered and constrained in the causal set literature.

The THPE is not falsified: it is *bounded*, and its local formulation is *closed*. We regard this set of negative results as a legitimate scientific product: it delimits the viable space of the hypothesis, documents methodological warnings of general utility (the non-convergence of ensemble samplers on degenerate posteriors; the necessity of coherence controls in model comparison), and leaves a falsifiable prediction with a verifiable time stamp.

9 Future testability

The distinctive prediction of the THPE — a crossing of $w(z) = -1$ whose chronology follows the star formation peak at $z \simeq 1.9$ — is *not* testable at present precision: the permitted couplings, Eq. (8), place the signature below the threshold of current geometric data.

The datasets that will permit the test are the final DESI release (expected around 2028–2029, on whatever schedule the collaboration establishes), the cosmological releases of Euclid [29] (2026–2030), and LSST at the Vera C. Rubin Observatory [30] from approximately 2027.

The public, dated registration of this prediction (repository commits of July 2026) precedes those data. Should dark energy prove dynamical with the predicted chronology, temporal priority is verifiable; should $w = -1$ be consolidated at sub-percent precision, the THPE will be refuted in its reformulations as well, and the authors will say so.

Code and data availability

The complete pipeline, the full correction history and the coherence controls are publicly available at <https://github.com/moimora1/THPE> (version 1.8): `THPE_fit_v18.py` (MCMC fit to DESI DR1/DR2 and Pantheon+), `THPE_dynesty_v1.py` (Bayesian evidence, BAO+SNe) and `THPE_dynesty_v2_cmb.py` (evidence with the CMB anchor). The **CHANGELOG** documents every version, including the errors detected and corrected during this work. Pantheon+ data are not redistributed; the scripts download them from the official **PantheonPlusSHOES/DataRelease** repository.

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